

Tiago Barbosa

AI Engineer | Generative AI, Agentic Systems & MLOps
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Porto, Portugal



[Linktree](#)

AI Engineer focused on building Generative AI systems that make it to production. At TensorOps, I work for international clients, turning ideas into deployed AI: on the adtech side, I rebuilt a programmatic bidder from the ground up (policy routing, log-normal bidding, profit optimization), improving performance by around 300% and turning a monthly loss into consistent profit; on the security side, I build agentic remediation systems with Generative AI and LangGraph. Background in Informatics and Computing Engineering (MSc, University of Porto, 18.1/20), with a dissertation on adversarial robustness in deep learning and time at TU Wien focused on Generative AI. AWS Certified Generative AI Developer, Professional.

Experience

AI Engineer

@TensorOps
Sep 2025 - Present

- Delivering Generative AI and adtech projects for **international clients**, owning implementation from conception to production deployment.
- Tappx** (adtech / SSP): Sole engineer on the bidder optimization project, **authoring ~95%** of the codebase. Designed and shipped an Algorithm Policy Router, log-normal bidding with 5% exploration, and a profit-maximizing bid optimization system.
- Improved bidder performance by ~300%**, turning a monthly loss into consistent profit (a **~\$12.7k profit** swing across the first quarter).
- Armis** (cybersecurity): Building agentic systems for automated remediation using Generative AI and LangGraph, orchestrating multi-step agent workflows for security operations.
- YieldAgent**: Core contributor to an early-stage product connecting AI agents to media platforms to generate ad campaigns.

Certifications

AWS Certified Generative AI Developer, Professional (AIP-C01) – Amazon Web Services, 2026

- Advanced certification validating the design, deployment, and operation of production-grade Generative AI solutions on AWS.
- Covers foundation model integration, RAG architectures, vector stores and knowledge bases, prompt management, guardrails, evaluation, security, and cost optimization.

Education

Faculty of Engineering of University of Porto

Porto, Portugal

Master's Degree in Informatics and Computing Engineering

Sep 2023 - Jul 2025

- Graduated with **18.1/20**
- Key Areas:** Artificial Intelligence, Deep Learning, AI Security / Adversarial ML, MLOps, Software Development (Large Scale), Distributed Systems (Large Scale), Enterprise Management and Project Management

Vienna University of Technology (TU Wien)

Vienna, Austria

Erasmus+ Exchange Programme

Sep 2024 - Feb 2025

- Key Areas:** Generative AI, Quantum Computing, Cryptocurrencies and Ubiquitous Computing (IoT)

- Graduated with **18/20**
- **Key Areas:** Artificial Intelligence, Parallel and Distributed Computing, Web Technologies, Computer Networks, Computer Security, Programming, Algorithms and Data Structures, Databases, Operating Systems, Algorithm Design and Compilers

Other:

Denmark TU (BEST Summer Course) about Computer Vision Copenhagen, Denmark Jul 2025

Boğaziçi University (BEST Summer Course) about Brand Management Istanbul, Turkey Jul 2024

Projects

Safe-DL: A Modular Framework for Secure Deep Learning (Master's dissertation)

@FEUP

Feb 2025 - Jul 2025

- Conceived and developed an original framework for evaluating and mitigating adversarial threats in deep learning systems. The solution integrates threat modelling, attack simulation, risk scoring, automated defence selection and audit-ready reporting. The project bridges deep learning with AI security and regulatory compliance (e.g., AI Act), enabling robust and transparent AI deployments.
- Graded **19/20**
- **Skills:** Python, PyTorch, Deep Learning, Adversarial Machine Learning, Cybersecurity, Risk Analysis, Framework Design, Autonomy, Research and Critical Thinking

Skills

Programming Languages: Python, Java, C, C++, JavaScript, TypeScript

AI & ML Frameworks: PyTorch, TensorFlow, OpenCV, LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, Ollama, scikit-learn, Keras, spaCy

Programming Paradigms & Concepts: Object-Oriented Programming (OOP), Functional Programming (FP), Concurrency, Parallelism, Multithreading, Multiprocessing

Cloud & DevOps Tools: AWS (SageMaker, Bedrock, Comprehend, Rekognition, Lambda, S3, EC2, IAM), Google Cloud Platform (Vertex AI, BigQuery, AI Platform, Cloud Functions), Microsoft Azure (Machine Learning, Cognitive Services, Functions, Blob Storage), Docker, Kubernetes, Git, GitHub Actions, CI/CD, Terraform, FastAPI

Databases & Search: SQL, NoSQL, Vector Databases, MySQL, ClickHouse

Key Topics & Interests: Artificial Intelligence, Deep Learning, Computer Vision, Generative AI, Large Language Models (LLM) Integration and Fine-Tuning, Retrieval-Augmented Generation (RAG), Agentic AI Systems, AI Security, Adversarial Machine Learning, Prompt Engineering, MLOps, MLflow, Model Deployment, API Integration, Data Engineering Pipelines, Adtech (Bid & Floor-Price Optimization)

Soft skills: Autonomy, Problem-Solving, Critical Thinking, Teamwork, Leadership, Communication, Adaptability, Time Management, Project Management, Creativity, Cross-Cultural Collaboration

Languages: Portuguese (Native), English (Fluent, C1)